

```
In [35]: import pandas as pd

data = {
    'Variable': [
        'Climate',
        'Housing',
        'Health',
        'Crime',
        'Transportation',
        'Education',
        'Arts',
        'Recreation',
        'Economy'
    ],

    'PC1': [
        0.190,
        0.544,
        0.782,
        0.365,
        0.585,
        0.394,
        0.985,
        0.520,
        0.142
    ],

    'PC2': [
        0.017,
        0.020,
        -0.605,
        0.294,
        0.085,
        -0.273,
        0.126,
        0.402,
        0.150
    ],

    'PC3': [
        0.207,
        0.204,
        0.144,
        0.585,
        0.234,
        0.027,
        -0.111,
        0.519,
        0.239
    ]
}

df = pd.DataFrame(data)

df.to_csv("pca_exercise2.csv", index=False)

print("Dataset created successfully!")
```

```
print("\nDataset:")
print(df)
```

Dataset created successfully!

Dataset:

	Variable	PC1	PC2	PC3
0	Climate	0.190	0.017	0.207
1	Housing	0.544	0.020	0.204
2	Health	0.782	-0.605	0.144
3	Crime	0.365	0.294	0.585
4	Transportation	0.585	0.085	0.234
5	Education	0.394	-0.273	0.027
6	Arts	0.985	0.126	-0.111
7	Recreation	0.520	0.402	0.519
8	Economy	0.142	0.150	0.239

```
In [37]: data = pd.read_csv("pca_exercise2.csv")

print("Dataset:")
print(data)

print("\nDataset Shape:")
print(data.shape)
```

Dataset:

	Variable	PC1	PC2	PC3
0	Climate	0.190	0.017	0.207
1	Housing	0.544	0.020	0.204
2	Health	0.782	-0.605	0.144
3	Crime	0.365	0.294	0.585
4	Transportation	0.585	0.085	0.234
5	Education	0.394	-0.273	0.027
6	Arts	0.985	0.126	-0.111
7	Recreation	0.520	0.402	0.519
8	Economy	0.142	0.150	0.239

Dataset Shape:

(9, 4)

```
In [39]: print("Variables:")
print(data['Variable'])

print("\nPC1 Loadings:")
print(data['PC1'])

print("\nPC2 Loadings:")
print(data['PC2'])

print("\nPC3 Loadings:")
print(data['PC3'])
```

Variables:

```
0      Climate
1      Housing
2      Health
3      Crime
4      Transportation
5      Education
6      Arts
7      Recreation
8      Economy
```

Name: Variable, dtype: object

PC1 Loadings:

```
0    0.190
1    0.544
2    0.782
3    0.365
4    0.585
5    0.394
6    0.985
7    0.520
8    0.142
```

Name: PC1, dtype: float64

PC2 Loadings:

```
0    0.017
1    0.020
2   -0.605
3    0.294
4    0.085
5   -0.273
6    0.126
7    0.402
8    0.150
```

Name: PC2, dtype: float64

PC3 Loadings:

```
0    0.207
1    0.204
2    0.144
3    0.585
4    0.234
5    0.027
6   -0.111
7    0.519
8    0.239
```

Name: PC3, dtype: float64

```
In [41]: index = data['PC1'].abs().idxmax()

variable = data.loc[index, 'Variable']
loading = data.loc[index, 'PC1']

print("PC1 Analysis")
print("Variable:", variable)
print("Loading:", loading)
```

```
PC1 Analysis
Variable: Arts
Loading: 0.985
```

```
In [43]: index = data['PC2'].abs().idxmax()

variable = data.loc[index, 'Variable']
loading = data.loc[index, 'PC2']

print("PC2 Analysis")
print("Variable:", variable)
print("Loading:", loading)
```

PC2 Analysis
Variable: Health
Loading: -0.605

```
In [45]: index = data['PC3'].abs().idxmax()

variable = data.loc[index, 'Variable']
loading = data.loc[index, 'PC3']

print("PC3 Analysis")
print("Variable:", variable)
print("Loading:", loading)
```

PC3 Analysis
Variable: Crime
Loading: 0.585

```
In [47]: for pc in ['PC1', 'PC2', 'PC3']:

    index = data[pc].abs().idxmax()

    variable = data.loc[index, 'Variable']
    loading = data.loc[index, pc]

    print(pc, "->", variable, "| Loading:", loading)
```

PC1 -> Arts | Loading: 0.985
PC2 -> Health | Loading: -0.605
PC3 -> Crime | Loading: 0.585

```
In [63]: import numpy as np

x = data[['PC1', 'PC2', 'PC3']].values

print("Shape of x:", x.shape)
print(x)
```

Shape of x: (9, 3)
[[0.19 0.017 0.207]
 [0.544 0.02 0.204]
 [0.782 -0.605 0.144]
 [0.365 0.294 0.585]
 [0.585 0.085 0.234]
 [0.394 -0.273 0.027]
 [0.985 0.126 -0.111]
 [0.52 0.402 0.519]
 [0.142 0.15 0.239]]

```
In [65]: import matplotlib.pyplot as plt

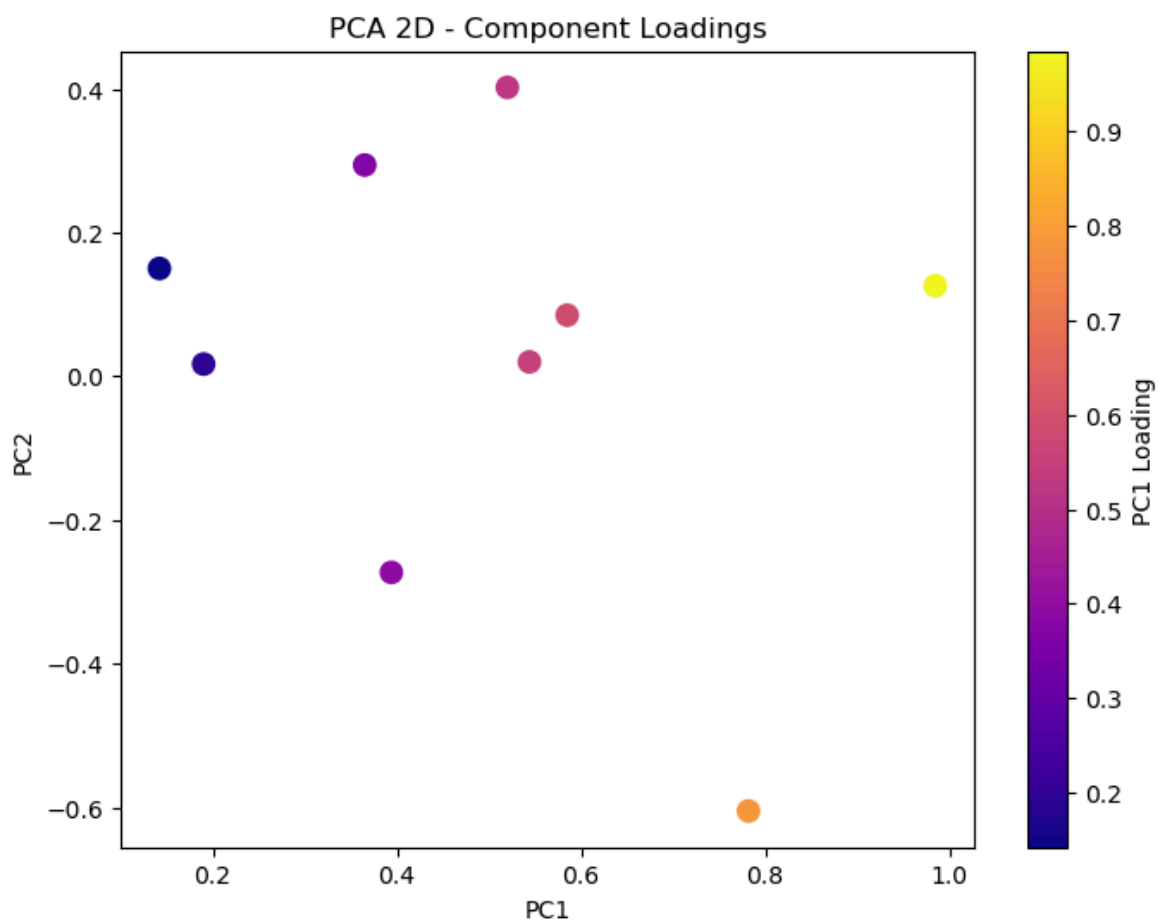
plt.figure(figsize=(8, 6))
```

```
plt.scatter(
    x[:, 0],
    x[:, 1],
    c=x[:, 0],
    cmap='plasma',
    s=80
)

plt.xlabel("PC1")
plt.ylabel("PC2")
plt.title("PCA 2D - Component Loadings")

plt.colorbar(label="PC1 Loading")

plt.show()
```



```
In [67]: import matplotlib.pyplot as plt

fig = plt.figure(figsize=(10, 8))

ax = fig.add_subplot(111, projection='3d')

scatter = ax.scatter(
    x[:, 0],
    x[:, 1],
    x[:, 2],
    c=x[:, 0],
    cmap='plasma',
    s=80
)

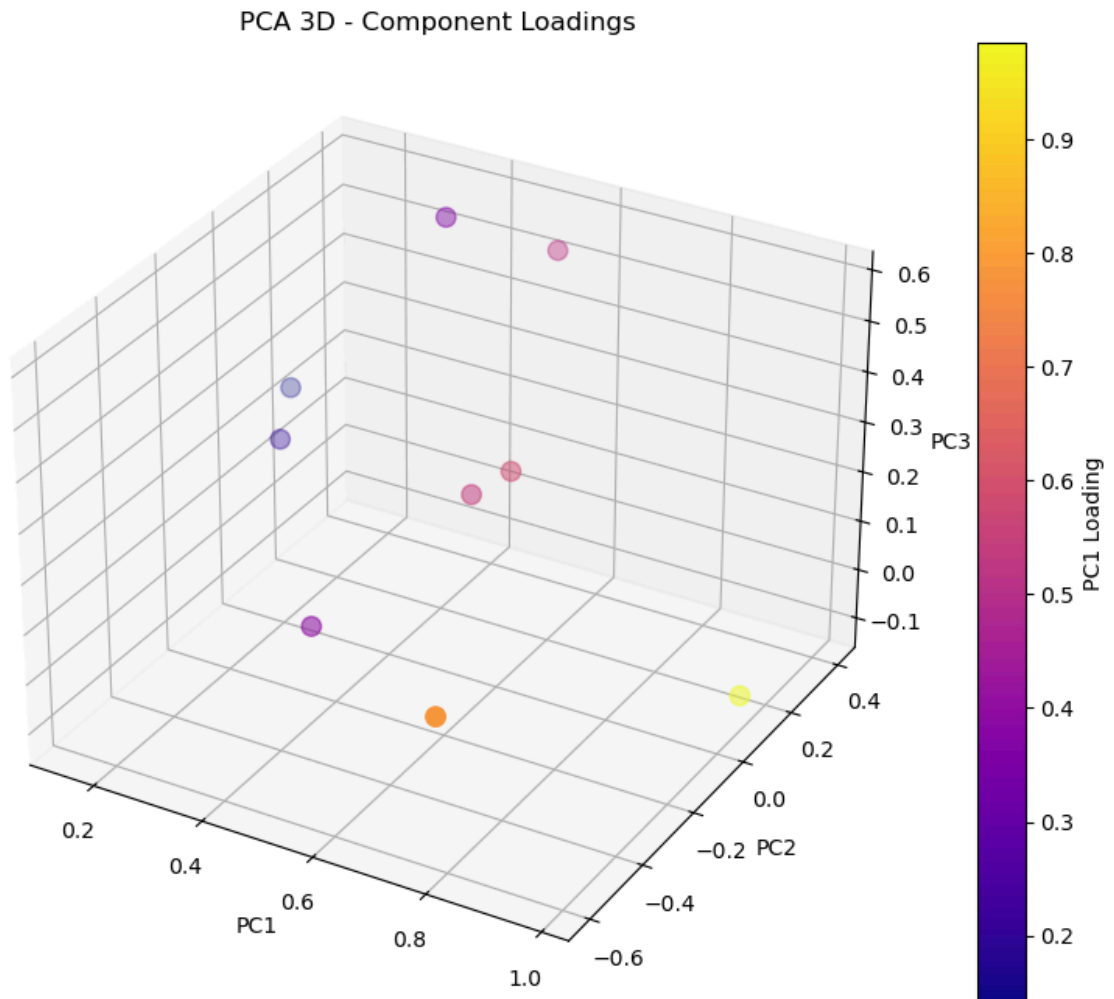
ax.set_xlabel("PC1")
```

```
ax.set_ylabel("PC2")
ax.set_zlabel("PC3")

ax.set_title("PCA 3D - Component Loadings")

plt.colorbar(
    scatter,
    label="PC1 Loading"
)

plt.show()
```



In []: